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The transcriptomic responses of the eastern oyster, *Crassostrea virginica*, to environmental conditions

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Abstract

Understanding the mechanisms by which organisms adapt to environmental conditions is a fundamental question for ecology and evolution. In this study, we evaluate changes in gene expression of a marine mollusc, the eastern oyster *Crassostrea virginica*, associated with the physico-chemical conditions and the levels of metals and other contaminants in their environment. The results indicate that transcript signatures can effectively disentangle the complex interactive gene expression responses to the environment and are also capable of disentangling the complex dynamic effects of environmental factors on gene expression. In this context, the mapping of environment to gene and gene to environment is reciprocal and mutually reinforcing. In general, the response of transcripts to the environment is driven by major factors known to affect oyster physiology such as temperature, pH, salinity, and dissolved oxygen, with pollutant levels playing a relatively small role, at least within the range of concentrations found in the studied oyster habitats. Further, the two environmental factors that dominate these effects (temperature and pH) interact in a dynamic and nonlinear fashion to impact gene expression. Transcriptomic data obtained in our study provide insights into the mechanisms of physiological responses to temperature and pH in oysters that are consistent with the known effects of these factors on physiological functions of ectotherms and indicate important linkages between transcriptomics and physiological outcomes. Should these linkages hold in further studies and in other organisms, they may provide a novel integrated approach for assessing the impacts of climate change, ocean acidification and anthropogenic contaminants on aquatic organisms via relatively inexpensive microarray platforms.

Keywords: adaptation, climate change, ecological genetics, ecotoxicology, genomics/proteomics, molluscs

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Introduction

Understanding the mechanisms by which organisms adapt to environmental challenges and the factors limiting their adaptive capacities has long been the 'holy grail' of biology. In marine environments, the natural stressors such as temperature, salinity, pH and dissolved

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oxygen often interact in complex ways leading to a complicated 'stress landscape', which demands physiological responses. The southeastern US estuarine systems show a remarkable degree of environmental variation so that creeks separated by a few kilometres may differ in the daily average of pH by as much as a full unit, salinities by as much as 20 ppt, dissolved oxygen by 20–50% and temperatures by 3–5 °C (http://cdmo.baruch.sc.edu/). In the recent past, this landscape has been further complicated by anthropogenic inputs of contaminants generating thousands of novel and often long-lasting compounds and by combustion of fossil fuels, which has contributed to global warming, ocean acidification and deposition of metals in estuaries impacting physiology and biochemistry of a wide range of species (Northeast States for Coordinated Air Use Management report 2003; Doney 2010). Thus, survival of marine organisms in the present-day estuaries requires acclimatization and/or adaptation to natural environmental variation and the additional anthropogenic pressures. In this report, we attempt to disentangle these complex interactions and partition the transcriptomic response of the oyster, C*rassostrea virginica*, to environmental variables.

The development of microarrays over the past decade has provided biologists with a vast repertoire of biosensors that allow testing mRNA expression of thousands of genes in response to environmental factors. To the extent that environmental stress modifies the level of gene expression, it should in principle be possible to dissect the complex patterns of gene expression and identify sensitive indicators of the effects of individual stressors on the organism's molecular machinery (cf. Steinberg *et al.* 2007 for a recent review). The importance of this technology in elucidating the response of marine organisms to environmental conditions has been documented (Gracey 2007; Hofmann & Place 2007, Evans & Somero 2008; Gracey *et al.* 2008; Place *et al.* 2008; O'Donnell *et al.* 2009). Mostly, these studies have focused upon the impacts of temperature (Hofmann & Place 2007; Place *et al.* 2008; O'Donnell *et al.* 2009) or salinity (Gracey 2007; Evans & Somero 2008; Gracey *et al.* 2008) on the transcriptome and identified genes and/or suites of genes in functional pathways that respond to these stressors.

In a previous study (Chapman *et al.* 2009), we examined the transcriptomic profiles of the oyster *Crassostrea virginica* for responses to land use practices in the surrounding watershed. This effort was motivated by the findings by Holland *et al.* (2004) that land use practices described by impervious surface and human population density could reasonably measure the degradation of the biological systems in tidal creeks. In this context, land use was an integrator of the diverse and complex stressors impacting tidal creeks. We reported that transcriptomic signatures tracked the land use patterns and could provide even finer discrimination of the relationship between watershed types and biological responses. The genes implicated in this response were largely components of energy conservation, protein synthesis and general metabolic machinery as opposed to 'stress' genes commonly used to assess environmental impacts such as metallothioneins, glutathione-S-transferase, heat shock proteins and cytochrome P-450. This finding was consistent with expectations based on studies of other marine and estuarine species (Gracey 2007; Hofmann & Place 2007; Evans & Somero 2008; Gracey *et al.* 2008, 2008; Place *et al.* 2008; O'Donnell *et al.* 2009). However, it is unlikely that the transcriptomic response of the organism to contamination can be divorced from the natural physico-chemical environment. As we will show below, this poses a significant analytical challenge for dissecting the interdependencies.

In this study, we extend our investigation of the transcriptional responses of oysters to the environment by analysing the relationships between gene expression and a suite of environmental (natural and anthropogenic) parameters. We employ the gene expression data used in our previous work (Chapman *et al.* 2009) and 160 additional arrays generated at the same time as those used in the previous work. The rationale for exclusion of these data from our previous work (Chapman *et al.* 2009) was that we focused on the impacts of human activities on oyster populations closest to those activities (first-order creeks, cf. Holland *et al.* 2004). Here, we analyse the extended data set to identify the specific environmental factors that may underpin the gene expression profiles. Specifically, we address the following questions within the context of chronic exposures to a wide variety of environmental conditions:

1 How does the transcriptome respond to variation in individual environmental stressors, and which genes and/or functional groups of genes are involved in the response?
2 How does the transcriptome respond to combinations of environmental stresses?
3 Do environmental stressors act independently or are there nonlinear and/or synergistic effects of multiple stressors on individual genes?

We assess the effects of individual environmental factors and their combinations on gene expression in three steps using machine learning tools that can model linear as well as nonlinear interactions among environmental variables. The first step maps transcript profiles to individual environmental stressors and detects complex interactions among genes to levels of environmen-

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tal factors using artificial neural networks (ANN's). ANNs have been widely employed in the analysis of complex data to discover patterns hidden from parametric statistics particularly in clinical applications (Bishop 2006; Haykin 1999; Khan *et al.* 2001; Cho & Won 2003; Linder *et al.* 2004; Dankbar *et al.* 2007; Sakai *et al.* 2007; Chu *et al.* 2008, and Das *et al.* 2008). In the second step, we develop and apply a linear clustering algorithm to generate combinations of environmental conditions and map gene expression to the resulting clusters. We evaluate the precision of these clusters using support vector machines (SVM) rather than logarithm of the odds scores typically employed in analysis of clustering algorithms. Once this clustering is accomplished, we perform ANN modelling and evaluate the robustness of the generated models using receiver operating characteristic (ROC) curves. Finally, we reverse the modelling approach to map stressors to the expression of individual genes and examine the relative contribution of environmental factors to individual gene expression. Machine learning tools are widely used in many scientific disciplines but their application to analysis of transcriptomic data in ecological settings is rather novel, and we hope this study will demonstrate the power of this approach in disentangling the complex patterns of gene–environment interactions.

Materials and methods

Sampling design

The details of the field sampling are given by Chapman *et al.* (2009). Briefly, eleven tidal creeks located from New Hanover County, NC, to Glynn County, GA, were sampled in June–August 2006 (Fig. 1). We aimed to collect equal numbers of oysters (*C. virginica*) from the first-, second- and third-order (cf. Holland *et al.* 2004) portions of each creek. This was not possible in some locations because second- or third-order creeks were not present or oysters did not inhabit second- or third-order creeks. Each creek was sampled on the ebb tide, approximately 2–3 h prior to low tide. Within each creek, 25 oysters were collected to analyse the transcriptome profiles and an additional sample of up to 12 oysters was taken for chemical analyses. Oysters were dissected in the field, and samples of gill and hepatopancreas stored in RNA*later*® (Ambion Inc., Austin, TX, USA).

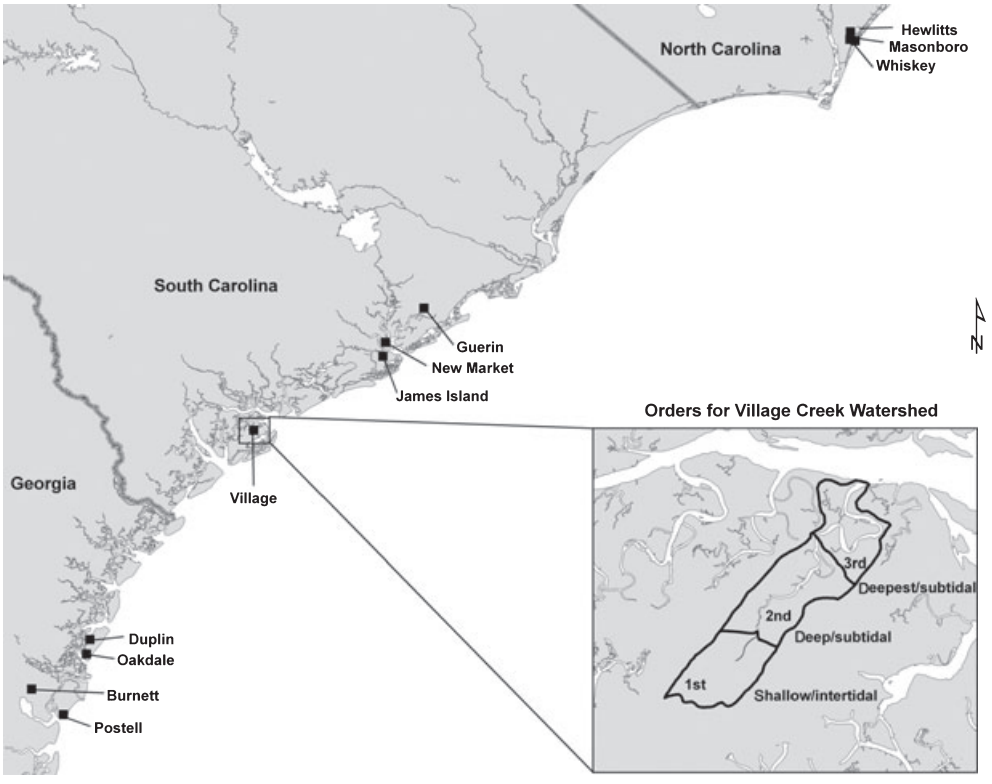


Fig. 1 Map of collection locations for oysters along the southeastern Coast of the US. The inset figure of Village Creek illustrates the assignment of creek order to location within creeks.

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Environmental and chemical analyses

Water quality data (temperature, dissolved oxygen, salinity, pH, turbidity, chlorophyll a, NH_4) were collected in bottom waters (0.3 m above bottom) using a YSI 6600 data logger adjacent to the site of oyster collection. These data measurements were made at 15-min intervals for two full tidal cycles (25 h) immediately preceding the collection. The average of these values over the recording period was used in the analysis of gene expression data.

The methodological details of the chemical analyses are presented by Sanger *et al.* (2008). Briefly, tissues from oysters collected at each study site were combined to a total mass of 15 g and extracted for contaminant analyses as described by Sanger *et al.* (2008). Elemental analysis was performed using inductively coupled plasma mass spectrometry (ICP-MS) and organic contaminants were measured by gas chromatography and mass spectrometry as appropriate.

RNA extraction and synthesis of labelled RNA target for microarray hybridizations

Total RNA was isolated from oyster tissues stabilized in RNA*later*® (Ambion Inc) using the RNeasy® Mini kits (Qiagen, Valencia, CA, USA) with an on-column DNAse (Qiagen) treatment. Extracted RNA was quantified by absorbance at 260 nm using a NanoDrop® ND-1000 Spectrophotometer (NanoDrop Technologies, Wilmington, DE). One microgram of total RNA was used to produce Cy3-labelled aminoallyl RNA (Cy3-aRNA) probe using the Amino Allyl MessageAmp™ II aRNA Amplification Kit (Ambion Inc) according to the manufacturer's instructions. Ten micrograms of Cy3-aRNA was diluted (1:3) in hybridization buffer [50% formamide, 2.4% SDS, 4× SSPE, 2.5× Denhardt's solution, and 2 μL of blocking solution (1 μg Cot-1 DNA and 1 μg poly (dA)]. The probe was then boiled for 1 min and incubated in the dark for 1 h at 50 °C prior to hybridization.

Microarray hybridization

The oyster cDNA microarray slides (Jenny *et al.* 2007) were prewashed with 0.2% SDS for 2 min, just-boiled Milli-Q water for 2 min, rinsed in 70% ethanol for 2 min and then dried. Subsequent steps were performed in the dark. Slides were prehybridized in a buffer (33.3% formamide, 1.6% SDS, 2× SSPE, 1.6× Denhardt's solution and 0.1 μM salmon sperm DNA) for 1 h at 50 °C. Slides were hybridized with Cy-3-aRNA for 16 h at 50 °C, rinsed in 2× SSC, 0.1% SDS and soaked in 0.2× SSC, 0.1% SDS for 15 min at room temperature, followed by a rinse in 0.2× SSC and sequential soaking in 0.2× SSC for 15 min, 0.1× SSC for 15 min and Milli-Q water for 5 min to remove carryover SDS. The microarrays were then dried and scanned with ScanArray™ Express and SpotArray software at 70 V photomultiplier tube gain and analysed with QuantArray software (Perkin Elmer, Boston, MA, USA). The microarray data are available at NCBI GEO Series GSE14793 (http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE14793).

Data analysis

Selection of variables. Tissue levels of metals and metalloids (As, Cd, Cu, Cr, Fe, Hg and Pb), organic contaminants (chloropyrifos, naphthalene, total DDT and total PAHs) and water quality measures (temperature, salinity, pH, turbidity, dissolved oxygen, chlorophyll a and ammonium levels) were selected for analysis. Tissue levels of inorganic and organic contaminants were considered a proxy for the environmental exposures to these pollutants. Measuring levels of the contaminants in oyster tissues rather than in the surrounding water and sediments has an advantage of integrating environmental exposures over the prolonged periods of time (potentially, the lifetime of oysters for some persistent pollutants) and providing physiologically relevant measures of individual exposure to these contaminants (cf. Macey *et al.* 2010). Henceforth, the above-mentioned tissue levels of chemicals and water quality parameters (a total of 18, Table 1) will be referred to as the environmental variables. Environmental variables used in this analysis were selected from a much larger suite of chemical and water quality measures reported by Sanger *et al.* (2008) based upon their weak or nonsignificant correlations with the others except for the tissue levels of Cu and Fe that were highly correlated with each other in the available data. The latter two variables were retained because both Cu and Fe are essential co-factors in a variety of biochemical processes. Different PCBs measured in the chemical analyses were excluded because in the present data, they were tightly correlated with chloropyrifos, which will stand in for all PCBs in the further statistical analysis. DDT and naphthalene, which are PAHs, were not correlated with the total PAHs levels and thus retained for the analysis.

The transcriptome response to variation in individual environmental stressors and identification of genes and/or functional groups of genes involved in the response. We mapped transcript profiles to individual contaminants using artificial neural networks (ANNs) as described elsewhere (Chapman *et al.* 2009). The analysis consists

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Table 1 Contaminant levels in oyster tissues in the surveyed creeks. Metal values are given in ppm dry weight, organics in ppb dry weight and chlorophyll A and ammonia concentrations (Chl-a and NH_4^+) are assessed per mg/L of creek water

Location	K means	As	Cd	Chl-a	Chlorpyrifos	Cr	Cu	DDT's	DO	Fe	Hg	Naphthalene	NH4	PAH	Pb	pH	Sal	Turb	Temp
Burnett Creek 1	4	4.67	1.006	41.802	0.9975	2	56.45	8.51	4.108	301.5	0.353	0	0.070	2161	0.418	7.332	20.627	27.387	30.882
Burnett Creek 2	4	5.92	1.115	17.571	0.4455	2.18	58.35	1.49	3.628	313.5	0.417	48.7	0.006	850.4	0.441	7.332	26.454	19.863	31.179
Duplin Creek 1	2	7.67	0.653	16.762	0	6.98	13.1	2.93	2.708	501	0.071	101	0.156	101	0.213	7.212	30.596	39.060	29.222
Duplin Creek 3	6	9.56	1.11	10.046	0.964	7.52	47.7	2.75	3.551	377	0.062	158	0.000	646	0.228	7.303	30.231	21.717	28.632
Guerin Creek 1	5	5.75	3.01	12.169	0	6.19	186	2.53	4.639	1020	0.142	0	0.013	0	0.752	7.235	22.402	15.742	28.961
Guerin Creek 2	1	9.75	3.66	5.251	0	7.17	562	4.6	4.794	615	0.225	141	0.025	141	0.350	7.326	22.718	4.694	28.664
Guerin Creek 3	1	10.8	3.02	5.553	0	9.14	473	4.12	4.552	1060	0.143	100	0.023	100	0.823	7.383	22.594	5.760	28.633
Hewlitts Creek 1	4	14.4	0.890	26.379	0.515	2.570	171.5	13.255	5.132	568	0.103	225	0.018	225	0.639	7.202	13.358	23.086	30.807
Hewlitts Creek 2	3	33.4	0.805	3.242	0	2.21	88.1	8.21	5.226	244	0.080	0	0.011	0	0.335	7.867	33.214	6.625	29.750
James Is Creek1	5	9.2	1.02	12.049	0	9.04	95.5	10.58	2.806	701	0.119	0	0.332	32.3	0.568	7.179	22.019	21.066	29.260
James Is Creek 2	6	10.5	1.21	10.679	0	8.01	59.7	11.11	4.098	551	0.086	95.7	0.000	127.6	0.540	7.302	25.410	11.907	29.760
James Is Creek 3	2	14.3	2	10.814	0	9.58	197	8.95	4.890	673	0.071	174	0.001	372.5	0.613	7.604	24.291	35.467	29.608
Masonboro	3	27.5	0.875	4.285	0.82	2.25	41.7	2.98	5.378	254	0.105	62.6	0.000	62.6	0.277	7.967	34.969	8.845	29.250
New Market Creek	4	12.4	1.45	8.561	1.59	2.18	110	20.84	3.516	480	0.107	92.9	0.649	718.2	1.580	7.613	18.621	34.686	29.370
Oakdale Creek 1	2	10.9	0.822	16.812	0	9.21	31.8	3.25	3.266	552	0.082	116	0.094	116	0.305	7.324	30.762	46.799	30.472
Postell Creek 1	4	9.35	1.09	28.124	0	2.39	73.6	4.22	3.459	407	0.145	161	0.365	161	0.300	7.306	23.017	32.268	30.303
Village Creek 1	2	8.65	0.881	19.934	0	6.35	56.4	8.88	4.297	299	0.102	46	0.008	46	0.309	7.495	32.121	20.735	31.585
Village Creek 2	2	13.3	2.04	10.027	0.961	10.7	82.6	2.73	3.493	340	0.065	73.8	0.024	73.8	0.281	7.421	32.244	16.897	31.559
Village Creek 3	2	14.7	2.46	10.995	0.878	8.84	68.4	3.02	5.170	370	0.070	68.3	0.015	68.3	0.267	7.700	31.810	38.725	30.772
Whiskey Creek 1	5	17.2	0.673	7.144	0	7.695	215	10.18	4.346	525.5	0.097	47.6	0.082	47.6	0.493	7.292	19.797	13.735	29.621
Whiskey Creek 2	3	21.5	0.402	6.692	0	7.47	121	8.15	4.594	347	0.091	34.1	0.003	34.1	0.416	7.769	30.835	11.622	29.461

Sal, Salinity in parts per thousand; Turb, turbidity in NSU units; Temp, temperature in degrees C. K-means indicates the cluster membership for indicated sampling locations (see Methods for details).

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of an initial filtering of the data using Bioconductor (Gentleman *et al.* 2004; R Development Core Team 2008) to normalize the data (variance-stabilized normalization) and eliminate genes, which did not exceed background levels on at least 10% of the slides.

In the present study, we have used several modifications of the ANN method. Briefly, we began the ANN analysis by mapping all genes that passed initial filtering to individual contaminants and water quality measures using five rounds of ANN training and randomly selecting 70% of the samples for each training session. The samples not used in the training session were used as a cross-validation set to test model robustness. The R^2 (correlations between observed values and those predicted by the ANN) were calculated for both training and cross-validation samples. The sensitivities of each gene to each environmental variable were then calculated using the model with the highest combined R^2 of the training and cross-validation sets. Sensitivities are a measure of the contribution of the gene to the explained variation. We then selected 250 genes with the highest sensitivities and conducted 20 rounds of ANN training using the same procedures outlined previously. The final treatment of the data in this fashion selected the genes in the top 250 that were unique to each environmental variable and conducted 20 training sessions as before.

The transcriptome responses to combinations of environmental stresses. We performed a K-means clustering of the environmental variables using WEKA version 3.6 (http://www.cs.waikato.ac.nz/ml/weka/), which clusters n objects into k partitions based upon multiple attributes. In the present case, we clustered sampling locations using the contaminant and water quality measures as attributes and examined the performance of the clustering using the support vector machine (SVM) method contained in WEKA. The SVM assesses misclassification rates in a cross-validation sample not used to train the SVM (cf. Vapnik 1999; Brown *et al.* 2000; Bishop 2006). We examined 2–10 clusters and found that six clusters gave the best correct classification percentage. Sampling locations were pooled as shown Tables 1 and 2. The individual samples from each location were then assigned to the appropriate cluster, and the entire ANN procedure was repeated as outlined previously, mapping genes to clusters in a pairwise fashion, e.g. cluster 1 vs. 2 and 1 vs. 3. As the cluster contained varying number of samples, we balanced the design by taking 70% of the samples from the smaller of the two compared clusters and drawing at random an equivalent sample size from the larger set. The observed and model-predicted cluster assignment for the cross-validation samples were analysed using receiver operating characteristic curves that are widely used in clinical diagnostics to assess model robustness (cf. Hanley & McNeil 1982; Bradley 1997; Braga & Oliveira 2003; Obuchowski *et al.* 2004).

Interactive effects of stressors on gene expression. To determine whether the effects of environmental stressors on individual genes are independent or interact nonlinearly, we mapped the environmental variables to individual gene expression profiles using ANNs. This is a

Table 2 Means of the environmental variables used to generate clusters via K-means of the locations listed in Table 1

Parameter	Full Data	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6
As	12.99	10.28	11.59	27.47	10.19	10.72	10.03
Cd	1.41	3.34	1.48	0.69	1.07	1.57	1.16
Chloro_a	14.15	5.40	14.22	4.74	24.80	10.45	10.36
Chlorpyrifos	0.35	0	0.31	0.27	0.68	0	0.48
Cr	6.01	8.16	8.61	3.98	2.32	7.64	7.77
Cu	135.47	517.5	74.88	83.6	106.9	165.5	53.7
DDT_Total	7.11	4.36	4.96	6.45	10.26	7.76	6.93
DissOxy	4.23	4.67	3.97	5.06	4.16	3.93	3.82
Fe	503.07	837.5	455.83	281.67	439.67	748.83	464
Hg	0.13	0.184	0.077	0.09	0.20	0.12	0.074
Naphthalene	89.58	120.5	96.57	32.23	125.43	15.87	126.85
NH_4	0.09	0.024	0.05	0.01	0.19	0.14	0
PAH_Total	286.79	120.5	129.6	32.23	723.43	26.63	386.8
Pb	0.49	0.57	0.33	0.34	0.67	0.60	0.38
pH	7.43	7.35	7.46	7.87	7.33	7.24	7.30
Salinity	25.52	22.66	30.30	33.01	19.24	21.41	27.82
Turbidity	21.81	5.23	32.95	9.03	26.73	16.85	16.81
WaterTemp	29.93	28.65	30.54	29.49	30.56	29.28	29.2

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generalization of linear regression methods commonly employed in analysis of microarray data (cf. Waters *et al.* 2003; Whitehead & Crawford 2006; Williams *et al.* 2007). Our approach has advantages because (i) it is not limited to linear combinations of the input (environmental) variables, (ii) estimation of equations is not constrained by sample size (degrees of freedom) and (iii) it does not assume independence of the gene expression. However, the limitation is that we cannot appeal to any statistical distribution to determine significance, as the estimation of equations is not constrained and must rely upon R^2 and cross-validation to examine model robustness. This is computational brute force, where all genes passing the initial filtering were examined and five models were run for each gene, but the sensitivities were calculated only for those genes whose model R^2 were in the top 250. In addition, we examined the effect of the two environmental variables with the highest sensitivities on selected genes using the graphic approach derived from Cannon & McKendry (2002). This approach clamps all but two input variables to their means and allows the remaining two to vary across the range of values contained in the data. The graphical representation is intended to illustrate the response of the output to changes in the two dominant input variables.

Results

The values for each environmental variable considered here are reported in Table 1. The concentration of metals and organics in the oyster tissues are below actionable levels set by EPA (www.epa.gov/osa). The total number of samples was 268 and 266 for gill and hepatopancreas tissues, respectively. The sample loss in hepatopancreas was because of poor RNA yields, poor hybridization, high background or absence of tissue contaminant data. A total of 2720 and 2860 genes passed the filtering criteria in gill and hepatopancreas, respectively.

Mapping the transcriptome to environmental stressors

Model characteristics. This analysis maps gene expression profiles to individual environmental stressors and the results are presented in Fig. 2a–e. The ANN models using all genes that passed the filtering criteria for gill and hepatopancreas (Fig. 2a,d) indicated reasonable accuracy (high R^2) of the transcripts' response to most individual environmental variables with the exception of naphthalene in both gill and hepatopancreas. However, with some exceptions, the cross-validation data yielded much lower R^2. In gill tissues, the training and cross-validation sets for As, total DDT, Pb and pH were not statistically different as the standard deviations in R^2 overlapped (Fig. 2a). In hepatopancreas, the training R^2 were generally higher than in gill tissues, but the standard deviations of R^2 rarely overlapped between the training and validation sets except in cases where the standard deviations of the training R^2 were quite large (e.g. for pH, Fig. 2d). In general, models of gene expression identified for hepatopancreas were superior to those from gill tissues (higher R^2 for the training sets), but less robust as the R^2 were much lower for the cross-validation sets than for the training sets.

Reducing the number of analysed genes to those with sensitivities in the top 250 (Fig. 2b,e) decreased the model R^2 in most cases, but the standard deviations increased. Further, the disparity between model R^2 and cross-validation R^2 diminished for most of the environmental variables. For example, in the total data set, the average model R^2 for Cd was about 0.8 with a cross-validation R^2 about 0.2. In the reduced data set, the model R^2 remained about the same, but the cross-validation set rose to about 0.5. Similar observations hold for most of the environmental variables in both tissues.

Further reduction in the gene set for ANN analysis to the genes unique to each of the environmental variables (Fig. 2c,f) yielded substantially reduced model R^2 values and universally poor cross-validation R^2 for hepatopancreas. In the gill tissues, the cross-validations for total DDT, Pb, pH and salinity are reasonably close to the model values, but otherwise the performance was not impressive. In the gill tissue analyses (Fig. 2a–c), the models predicting metal concentrations are generally superior to other environmental variables (e.g. organic pollutants, nutrients and water quality measures). In hepatopancreas (Fig. 2d–e), the R^2 values were similar for various environmental measures.

Informative genes. The genes that were ranked in the top 250 based upon sensitivity for each of the environmental parameters are listed in Tables S1 and S2 (Supporting information). We have combined sequences in Tables S1–S4 (Supporting information) that received the same gene annotation (e.g. actin), but were not combined into a contig by CAP 3. These may represent noncontiguous segments of the same gene or different loci. Hence, the number of genes reported in these tables is less than the number of sequences actually used in the analysis. These data indicate that several metabolic systems notably respond to environmental variation and the involved genes are similar in gill and hepatopancreas. First, the entire suite of genes of the electron transport chain is an important discriminator for the levels of metals, organic pollutants and nutrients in both gill and hepatopancreas. It includes the mitochondrially encoded

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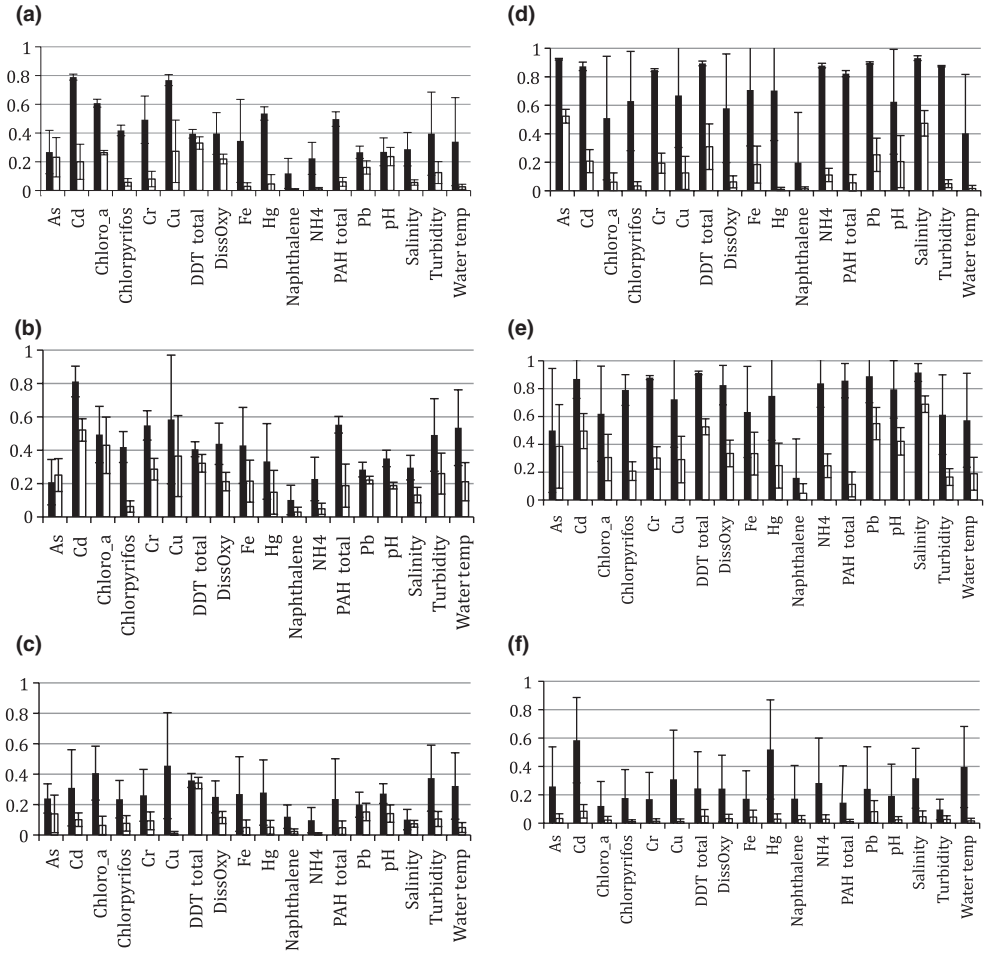


Fig. 2 R2 (*y*-axis) for training (dark bars) and cross-validation (light bars) data sets and standard deviations mapping gene expression to the indicated environmental variable. (a) Gill all genes, (b) Gill 250 most informative genes, (c) Gill genes unique to indicated environmental variable, (d) Hepatopancreas all genes, (e) Hepatopancreas 250 most informative genes, (f) Hepatopancreas genes unique to indicated environmental variable.

cytochrome *b*, cytochrome *c* oxidase I, II and III, NADH dehydrogenase subunits 1–6, and ATPase subunits 6 and 8 (a total of 11 of the 12 mtDNA-encoded electron transport proteins). In addition, nuclear genes involved in electron transport were also important including cytochrome *c* oxidase IV, succinate dehydrogenase, malate dehydrogenase (mitochondrial and cytosolic), ubiquinone-NADH dehydrogenases as well as DNA-directed RNA polymerase II-H, which is essential for the translation of mitochondrially encoded genes (Asin-Cayuela & Gustafsson 2007). Transcription factors TFAM, TFB1M and TFB2M, which are essential for mitochondrial transcription, were not included in the microarray and thus not analysed.

In addition, a host of ribosomal proteins were important discriminators of at least one environmental variable, more notably in hepatopancreas than gill (67 vs. 18 genes, respectively). Elongation factors, eukaryotic translation initiation factors (eIF), RNA helicases (DEAH 35 and other dead box proteins) and a number of aminoacyl synthetases necessary for mRNA processing were also associated with mRNA expression for ribosomal proteins.

A third set of metabolic processes important in discriminating the levels of environmental stresses includes a suite of genes involved in the regulation of cell volume and growth including genes in the focal adhesion complex such as paxillin, caveolin, c-jun, cyclin b, cyclophilin and cell cycle proteins that regulate cell growth and volume via actin polymerization and depolymerization including actin, actin-related protein 2–3 complexes, actin depolymerization factor 3, profilin, alpha tubulin, calpolin, calponin/transgelin, thymosin beta and rho-GTPase (Table S1 and S2, Supporting information), Growth factors, which regulate activation of this cascade including ras, myc, egf, tgf-beta and

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lps-induced tgf-alpha (Table S1 and S2, Supporting information), were also important discriminators of the environmental parameters.

A number of genes involved in protein stabilization (including chaperonins and heat shock proteins, or Hsps) and DNA stabilization (histones) were also important discriminators. The histones are also involved in immune response (Patat *et al.* 2004) while the chaperonins and Hsps are well-known indicators of environmental stress. An additional notable group of gene discriminators includes effectors or targets of Ca^{2+}signalling and homoeostasis such as annexins, calmodulin, ef-hand and carbonic anhydrase (the latter governing bicarbonate flux and regulating intra- and extracellular pH). In addition to the functions of known genes in the context of environmental response, it is equally important to note that many genes of unknown function are responsive to environmental conditions (Table S1 and S2, Supporting information). Finally, many of the genes typically investigated for environmental stress including cytochrome P-450, P-glycoprotein, superoxide dismutase and metallothioneins are conspicuously absent from these lists, except for MT-1, which was important only in response to Hg.

Mapping gene expression to environmental clusters

K-means clustering. The intent of this analysis was to determine whether simple tools could identify combinations of environmental variables that provide addition insights into transcriptional response. The K-means clustering of measured environmental variables identified six groupings as solutions, which minimized misclassification using the SVM. The means of the variables for each cluster are presented in Table 2. The ANN models of gene expression profiles for cluster comparisons could be presented as above; however, to provide a clearer picture of the results, we compared the clusters pairwise using ROC curves to assess the precision of classification.

The results of mapping genes to the K-means clusters are presented in Fig. 3a–c. Using all genes that passed initial filtering to compare clusters (Fig. 3a), 15 of the total 30 comparisons did not differ from random expectations and 11 of these involved comparisons of hepatopancreas tissues. Using the 250 genes with the highest sensitivities (Fig. 3b), only 5 of 30 comparisons were not significantly different from random expectations and all of these involved comparisons of gill tissues. In four of these cases, the AUCs are significantly inferior to the results in Fig. 3a. However, 23 of the 30 total comparisons in 3b significantly improved in the discrimination of the clusters and 14 of these were comparisons of hepatopancreas tissues. Using the genes whose sensitivities were in the top 250 for at least five comparisons (3c), five comparisons were not statistically significant, 15 comparisons were significantly better than those in Figs 2 and 3A were significantly worse. Of the 15 cases in which discrimination of the clusters was improved, 11 were comparisons of gene expression in hepatopancreas. Thus, a reduction in the number of genes used in the discrimination algorithm results in improved classification precision and the improvement is more noticeable in hepatopancreas tissues than in gill.

Certain aspects of the comparisons of clusters in Fig. 3 are worth noting. As is typical of clustering algorithms, there is no obvious reason why most of the clusters appear, but two of the clusters in Table 2 (i.e. 4 and 6) differ from the others in elevated PAH levels. In particular, cluster 4, which is composed of locations with the highest PAH levels (Tables 1 and 2), is significantly discriminated from nearly all other clusters (Fig. 3) including cluster 6, which has the next highest PAH levels. Cluster 3, which is characterized by locations with high pH, low PAHs and low Fe, is strongly differentiated from all other clusters, especially in reduced gene sets illustrated in Fig. 3b,c. Cluster 1 contains sites enriched in Cu and Fe and is easily distinguished from all other clusters except cluster 5 that also contains sites with high Cu and Fe levels. These clusters are readily distinguished in the reduced gene set used in Fig. 3c, in the reduced gene set from hepatopancreas in Fig. 3b and in the total gene set for gill (Fig. 3a).

Informative genes. The genes that provide substantial discrimination of the K-means clusters are presented in Tables S3 and S4 (Supporting information). For the most part, these gene lists are identical to those in Tables S1 and S2 (Supporting information) including a large number of ribosomal proteins, genes related to bioenergetics including the electron transport chain, as well as genes involved in protein synthesis and growth. Thus, the clustering methods did not provide much additional information on specific genes that could not be discerned from the mapping of genes to individual environmental variables.

Mapping environmental variables to individual genes

This effort was intended to identify which of the environmental variables were most important for differential gene expression. Environmental pH and temperature were by far the leading environmental factors governing gene expression patterns with minor contributions of salinity and dissolved oxygen

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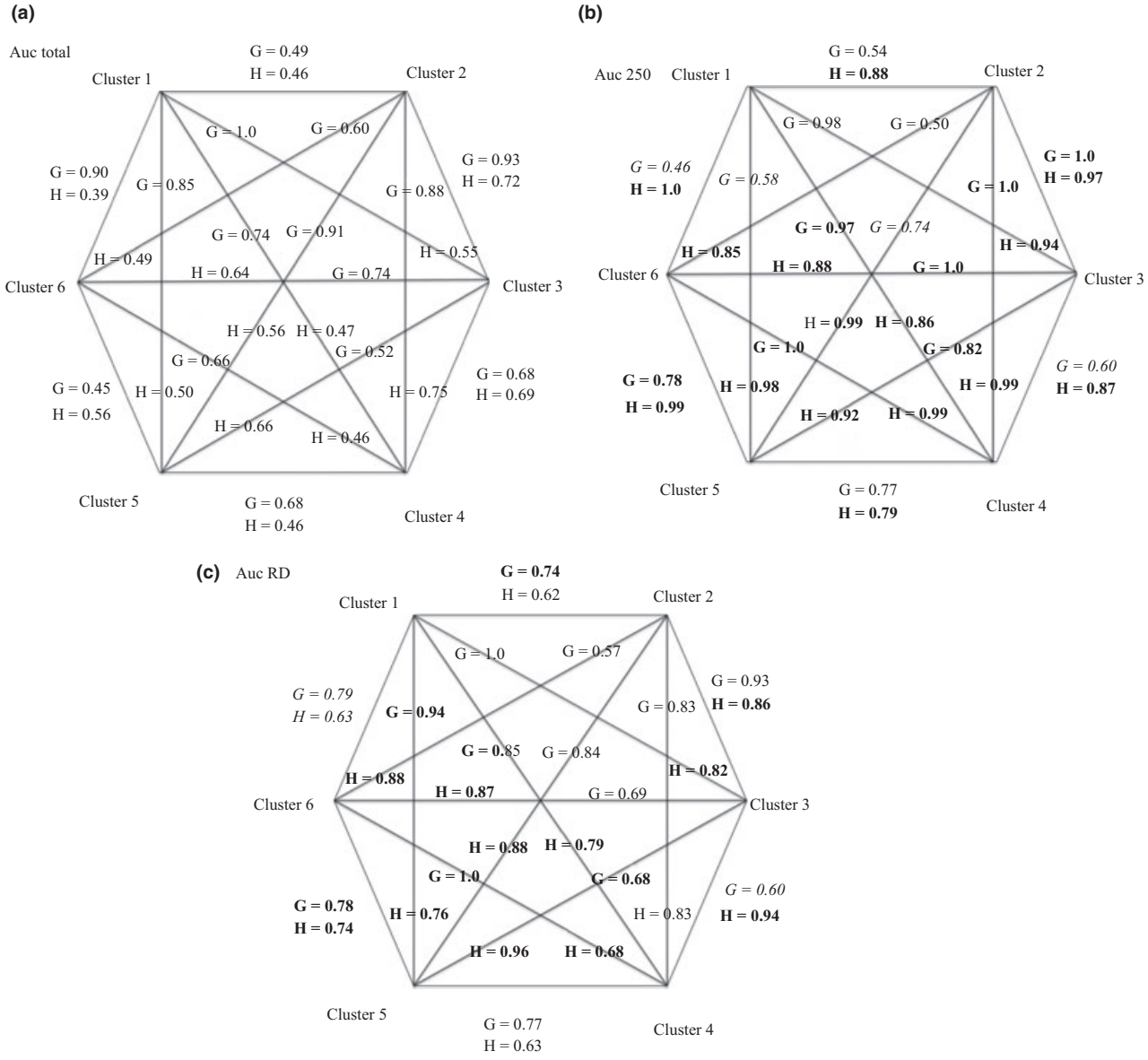


Fig. 3 The results of the receiver operating characteristic (ROC) curve analysis mapping genes to Kmeans clusters of the environmental variables. The values presented are the areas under the ROC curves and G = Gill tissues, H = hepatopancreas. (a) are the results for mapping all genes that passed initial filtering, (b) the 250 genes with the highest sensitivity of each comparison and (c) genes that were in the top 250 for at least five comparisons. Values greater than 0.64 were significant different from random expectations, values in bold (b,c) were significantly higher than in (a), and values in italic significantly lower.

(Fig. 4a,b). Notably, metals and organic contaminants played minimal roles in explaining the variation in gene expression. In Tables S5 and S6 (Supporting information), we present the average sensitivities for each environmental variable to each gene along with the average R^2 for the models and cross-validations (CV). Notably, gene expression in gill was far more predictable as model R^2 ranged between 0.77 and 0.9, with CV R^2 ranging between 0.58 and 0.94 while the hepatopancreas model R^2 ranged from 0.10 to 0.49 with CV R^2 ranging between 0.002 and 0.07.

As pH and temperature were the dominant variables, we examined the response of selected genes to these physical factors. We initially chose Hsp70 expression owing to its well-known response to environmental stress. The results for various ANN models are presented in Fig. 5. These models clearly indicate that Hsp70 is up-regulated with increasing temperature and decreasing pH, but the details vary among models. Fig. 5b–d suggests some sort of plateau effect at high temperature and low pH while Fig. 5A does not. Perhaps more importantly, all of the models in Fig. 5 indi-

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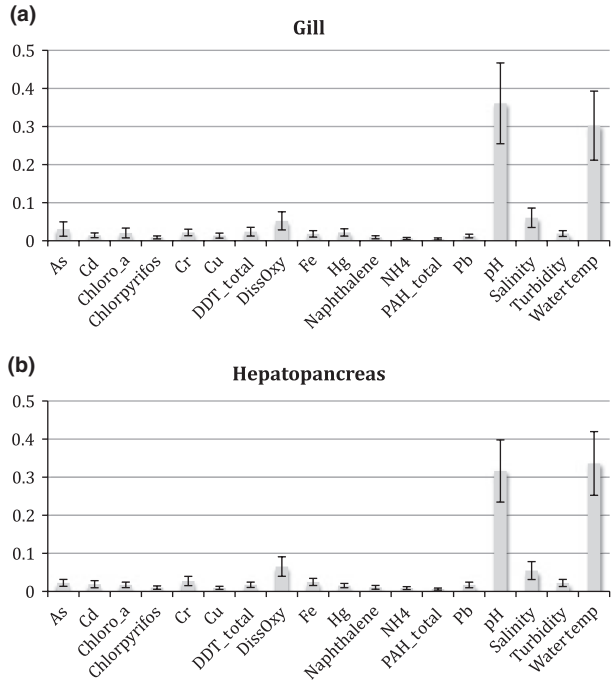


Fig. 4 Relative sensitivity (y-axis) of the genes used to generate Fig 3C to various environmental variables. The bars indicate the average sensitivity of all genes to the individual variables indicated on the *x*-axis for Gill (a) and hepatopancreas tissues (b). The error bars indicate the standard deviations of the sensitivities across all selected genes.

cate that temperature is a more important determinant of Hsp70 expression than pH. A common theme in all of these models is that at high pH, the impact of temperature is virtually linear, while at low pH, models B, C and to some extent D suggest a departure from strict linearity.

Other genes selected for this detailed examination were peroxinectin (Px), NADH dehydrogenase subunit 2 (ND-2), ribosomal protein L7 (RPL7) and cytochrome *b* (cytB) (Fig. 6). The data show that elevated gene expression levels of Px and ND-2 are predicted at low pH and high temperatures and are mostly influenced by the temperature (Fig 6a,b). RPL7 and cytB are expected to be down-regulated at lower temperatures and higher pH, and pH seems to be a more important factor driving expression of these genes than temperature (Fig. 6c,d). The predictions also suggest substantial nonlinearity and plateau effects in gene responses to pH and temperature. The data presented in Fig. 6 are the predictions of one of the five models generated for each gene and represent only the general response of each gene to the environmental variables. Other models differed in details of the response surface, but not in the general shape.

Finally, we also assessed the linear correlations of all genes that passed the filtering criteria to pH and temperature in gill and hepatopancreas tissues (Fig. 7a,b). The x coordinate is the correlation of the expression level of a given gene to pH and the y coordinate the expression for the same gene correlated to temperature. In gills, there is a strong negative correlation, suggesting that genes that are up-regulated by higher temperatures are also up-regulated by lower pH and *vice versa* (Fig. 7a). Further, the extreme ends of the distribution seem to be overpopulated, i.e. more genes are strongly associated with pH and temperature than those in the centre. The same plots for hepatopancreas data (Fig. 7b) show a very different pattern where the extremes are decidedly underrepresented and the lower right quadrant (negatively associated with pH and positively associated with temperature) seems to be overrepresented.

Discussion

Our study provides important insights into the fundamental questions concerning the patterns of transcriptomic responses to environmental variation and the potential nonlinear or synergistic interactions of the environmental stressors on individual genes and groups of genes. We will discuss these insights in detail below; however, it is important to mention some caveats of the approach used in this study. Functional interpretations of transcriptomic data typically assume that changes in the mRNA expression translate into changes in the proteome and subsequently metabolic processes. While correlation between mRNA and protein expression may not hold true for some genes, at the global transcriptomic level, a change in the gene expression profile is reflected in the proteome as recently shown by Newman *et al.* (2006) in yeast where 87% of the protein levels were correlated with their mRNA expression levels. More studies of the global proteomic vs. transcriptomic changes are needed to further test the robustness of such correlations; however, our current level of understanding of transcriptional regulation in stress response suggests that this is a feasible working hypothesis. The relationship between gene expression and the metabolic phenotype has also been experimentally validated. Phue *et al.* (2005) demonstrated that *E. coli* strains that differed in glucose metabolism also differed in the expression of genes related to glucose metabolism, and Lloyd & Zakhleniuk (2004) reported a similar observation in *Arabidopsis*. These studies were focused on the previously established metabolic differences between selected strains and did not analyse changes in expression among genes involved in other metabolic pathways. Nonetheless, they provide reasonable support for the coupling between gene expression and metabolic outcomes.

Two other caveats also deserve comment. First, we have averaged environmental variables (temperature,

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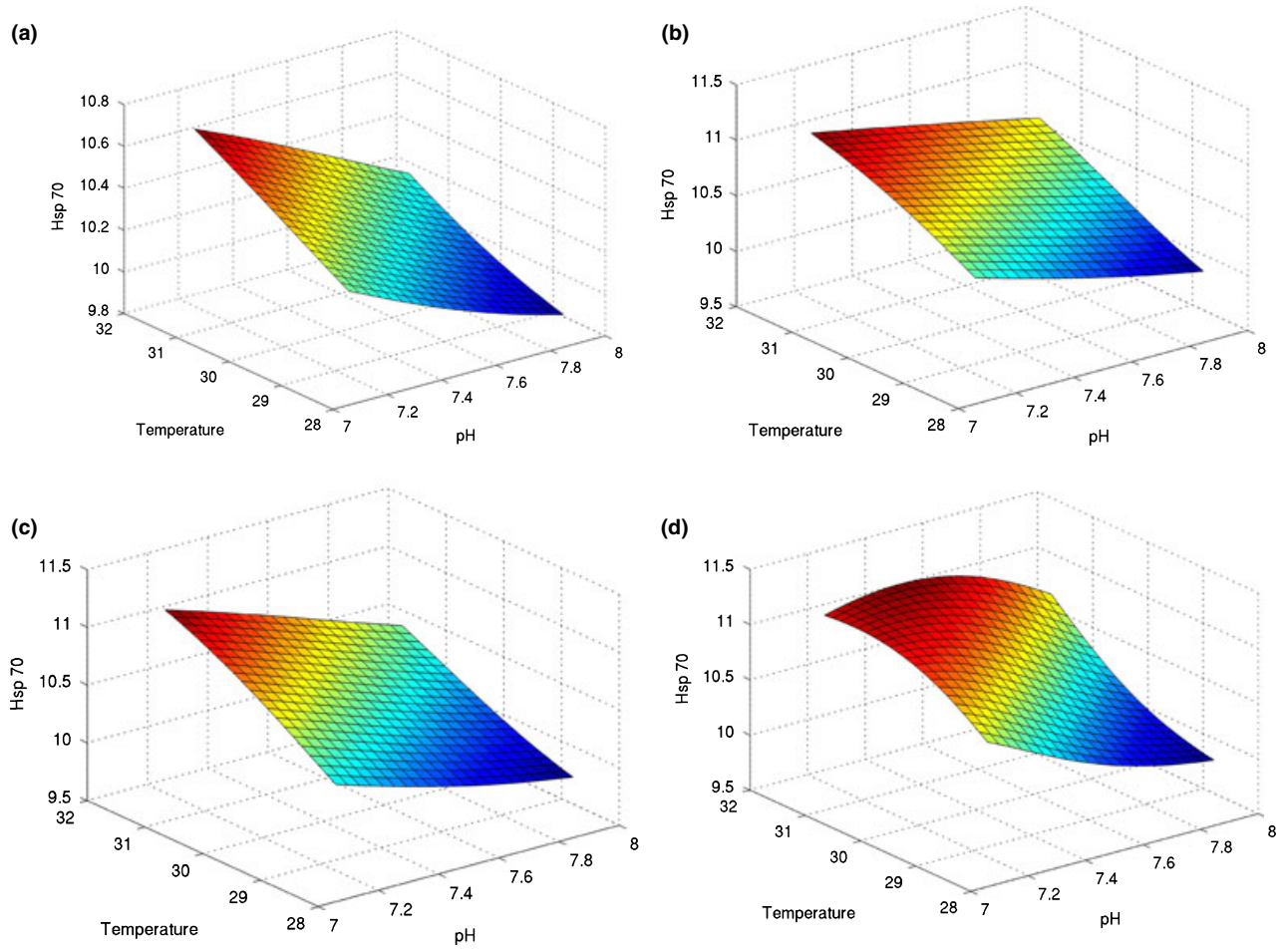


Fig. 5 Predicted expression of HSP 70 in gill from four ANN models allowing temperature and pH to vary across the ranged measured in the field and holding all other variables to their mean values. Gene expression values are in log base 2.

salinity, pH, etc.) over a 24-h period for each site so that our analysis does not take into account the potential site-to-site differences in environmental variability. Because tidal cycles can affect gene expression (Evans & Somero 2008; Gracey *et al.* 2008), we have taken steps to minimize the impact by sampling sites at the same tidal interval. Future studies outside the scope of this work could examine the effects of the diurnal or seasonal environmental variability within each site on the transcription profiles of oysters. Second, we have taken data from the chemical analysis of pooled oysters from the same site (but not the same individuals) as those used in the microarray studies. This pooling was necessary to generate sufficient biomass for the chemical analyses. All experimental oysters were collected within 1 m of each other minimizing the potential effects of microhabitat. In this study, we found no major influence of pollutants on gene expression and thus a more detailed examination of individual oysters was not deemed necessary.

How does the transcriptome respond to variation of individual environmental stressors, and which genes and/or functional groups of genes are involved in the response?

Methodological considerations. To address this question, we mapped transcriptomic responses to the levels of environmental variables. It is worth noting that nearly all of the reported model R^2 in this analysis (Fig. 2) would be considered statistically significant in linear statistics, except perhaps for naphthalene. All of the CV R^2 for physical variables (pH, temperature, turbidity, salinity and DO) and the metals (but not for most of the organic contaminants) would also be significant. Reduction in the number of genes used to predict environmental variables to the 250 with the highest sensitivities came at a small cost in terms of model precision (model R^2), but with an increase in robustness as evident in an increase in CV R^2. This observation is consistent with the curse of dimensionality (cf. Chapman

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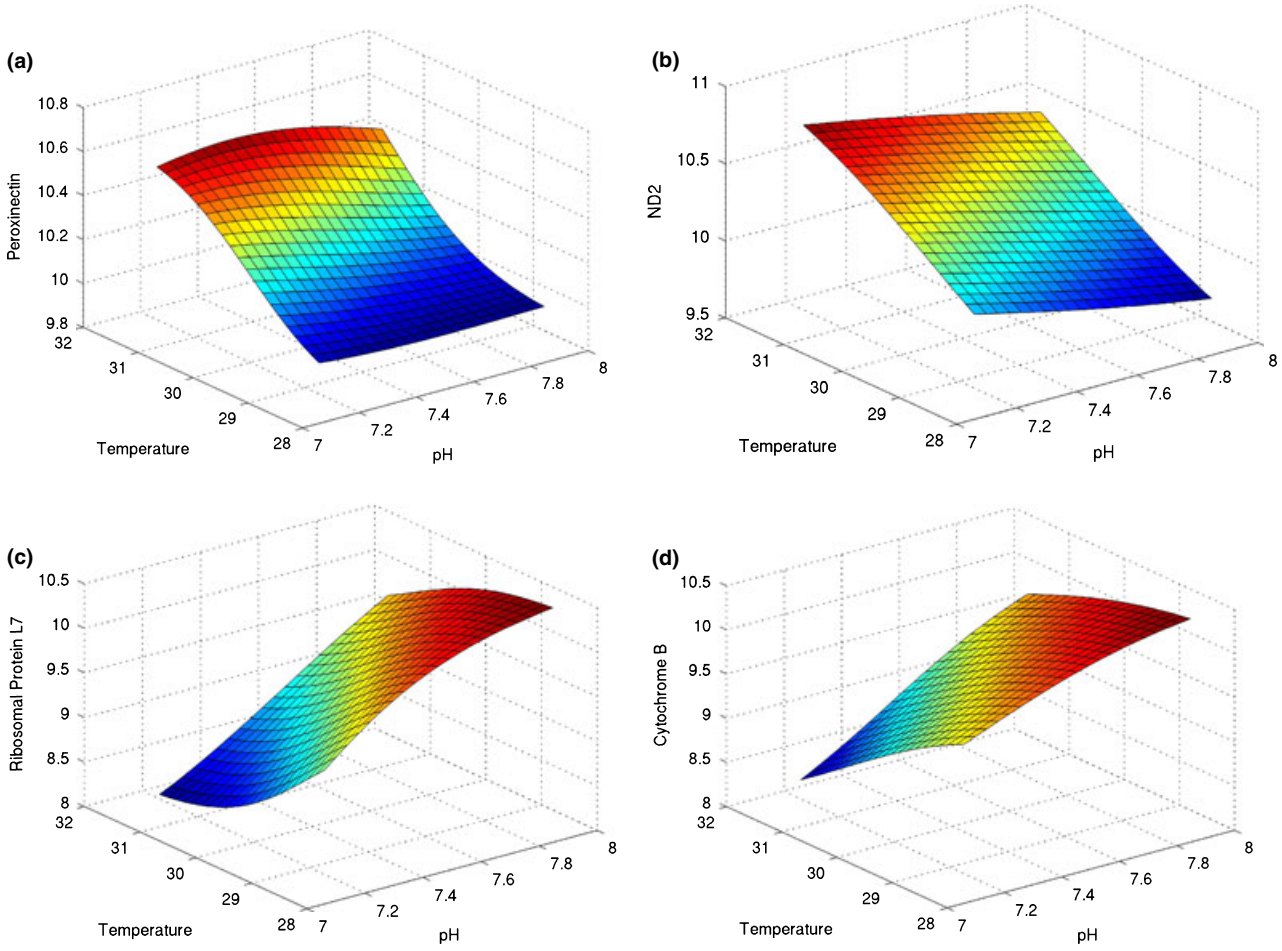


Fig. 6 Predicted expression of (a) peroxinectin, (b) ND 2, (c) ribosomal protein L7 and (d) cytochrome B in gill from ANN models allowing temperature and pH to vary across the ranged measured in the field and holding all other variables to their mean values. Gene expression values are in log base 2.

et al. 2009). Taking some licence with the precise language of mathematics, these observations suggest that using the full set of gene expression data permits the machine learning tool to 'memorize' the input records at the expense of making robust predictions of CV records (Fig 2a,d). Reducing the dimensionality of the problem (Fig. 2b,e) may sacrifice a modest level of precision for a substantial gain in robustness.

Data interpretation. The absence of a robust transcriptomic response to PAHs may stem from the fact that oysters do not have a recognizable aromatic hydrocarbon receptor. This receptor mediates most of the toxic effects of PAHs, at least in vertebrates (Brown *et al.* 1997). We did not find elevated levels of PAHs in oyster tissues relative to the environmental levels (data not shown). In contrast, transcriptomic responses to DDT and to a lesser extent chloropyrifos, which do not require aromatic hydrocarbon receptor, generated reliable and robust models. While the physiological effects of these pesticides are not well known in molluscs, our data show that they elicit a transcriptomic response at levels well below EPA actionable levels in a nontarget species such as oysters (Fig. 2b,e). Notably, among the metals, only Hg elicited a response from metallothioneins, whereas based on earlier laboratory studies, one might have expected a response to other metals, especially cadmium (Cd) (Butler & Roesijadi 2000; Jenny *et al.* 2004; Cherkasov *et al.* 2006 and references therein). Possibly, this reflects the fact that the tissue levels of Cd found in the present study are far below those used in laboratory challenges and thus might not be sufficient to stimulate expression of metallothioneins. However, we observed a general response of glutathione-S-transferase to metals (among others factors), which was especially pronounced for Cu and Fe. This suggests that glutathione-S-transferase expression may be a more sensitive indicator of the metal-induced

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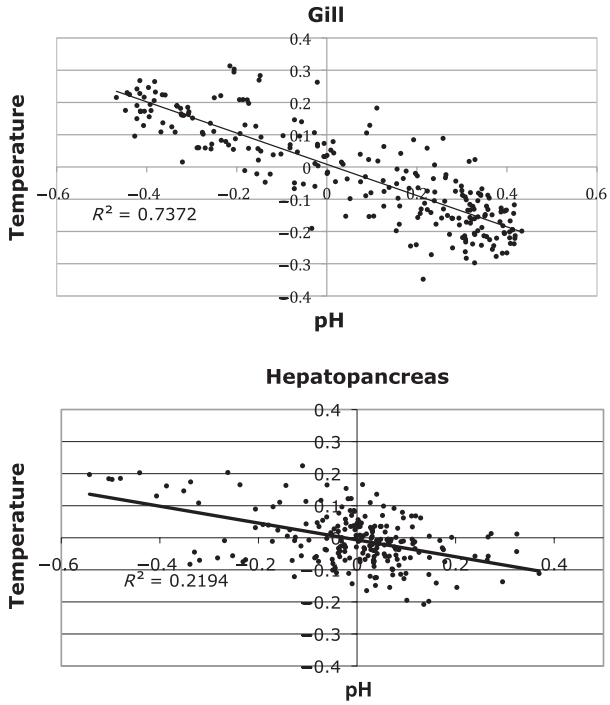


Fig. 7 Correlations of gene expression with pH and temperature for genes with the top 250 R2 from Fig 4. The point coordinates are x = correlation of expression of a given gene to pH and y = correlation of expression of the same gene to temperature.

stress than metallothioneins, particularly for the redox-active metals that can induce formation of reactive oxygen species via Fenton chemistry.

Our data also show a shift in β-actin expression in response to pollutants, suggesting that β-actin used for baseline corrections in qPCR in previous studies (e.g. Ivanina *et al.* 2008 and references therein) may not be always reliable as a housekeeping gene. This does not invalidate the results of these previous studies, but rather reinforces them. The shifting actin baseline would make the conclusions about the relative expression of other genes standardized to β-actin more conservative and underestimate the change in the expression of target genes. Relatively small expression changes may have been overlooked or submerged into the variance owing to a shifing β-actin baseline. In short, the conclusions of Ivanina *et al.* (2008) about the transcriptional response of stress proteins to metal challenge are likely underestimated.

To determine whether specific gene 'biomarkers' could accurately predict exposures to particular environmental stressors, we reduced the analysed set of genes to those uniquely responsive to each environmental variable; however, this led to a general decline in model and CV R^2 values. These results are disappointing, but enlightening, suggesting that the unique gene expression profiles must be embedded within general stress responses (i.e. genes included in Fig. 2b,e) to accurately predict exposure profiles. As such, the search for the magic bullet in the transcriptome reflecting exposure to specific environmental variables is likely to be context sensitive. We gain a great deal of information on the biological impact of specific environmental parameters by placing 'unique' responsive elements within the context of general response elements.

Among the functional groups of genes involved in response to environmental conditions, four groups stand out in Tables S1 and S2 (Supporting information): the electron transport chain proteins and mitochondrial ATP synthase; genes involved in protein synthesis (ribosomal proteins, translation initiation and elongation factors); cell growth (the focal adhesion complex); and cell shape and motility-related genes (including actins). We recognize that these processes cannot be divorced from each other and are part of a concerted activity of the organism coping with the environmental pressures. However, the data to this point suggest that gene expression in hepatopancreas is a more reliable predictor of environmental stress than expression levels in gill as the R^2 for both model and CV data sets for hepatopancreas generally outperform those from gill tissues.

Analysis of the transcriptome responses to combinations of environmental stresses and their potential nonlinear and synergistic effects on individual genes

Methodological considerations. The algorithms used to construct the clusters of environmental variables presented in Fig. 3 are based upon Euclidean geometry and do not consider the highly likely prospect that individual environmental variables will have different and likely nonlinear impacts on the transcriptome (as we will demonstrate below). Nonetheless, even this simplistic clustering offers significant insights. In particular, reducing the dimensionality of the inputs (Fig. 3b,c vs. 3a) results in a greater precision of classification as evident in the ROC curve values. However, reducing the set of considered genes to those that are general responders (Fig. 3c) does not perform quite as well as gene sets including general and specific responders (Fig. 3b). This reinforces the above-stated conclusions and strongly advocates for incorporation of general and specific gene responses to evaluate the impact of environmental conditions on biological systems.

The approach using K-means clusters is an unsupervised learning tool and does not consider the fact that some variables are more important than others. We recognize that there are metrics to consider these

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differences such as the ERMQ index of Long *et al.* (1995), but these indices create a number of analytical problems, particularly in application of machine learning approaches (cf. Montano & Palmer 2003). They are discrete, noncontinuous and not binary and while such indices are appealing from a heuristic point of view, it remains unclear how to manage or interpret the output.

The application of K-means clustering to generate grouping of environmental variables is also context sensitive in that it applies only to the data used in each specific analysis and is unlikely to permit a generalization beyond that data set. Therefore, we do not encourage this approach unless it can be shown that such clusters are generalization of the environment, which is unlikely to be true in most real-life scenarios.

The second portion of the analysis, which represents a generalization of the linear approaches used in most microarray studies, is perhaps most attractive in terms of identifying the interactive effects of environmental variables on gene expression. While it would be ideal to map nonlinear interactions of the environmental variables to collective responses of the gene expression profiles, at present such mapping is impossible because, to the best of our knowledge, no appropriate algorithms exist that would accomplish this. Given this limitation, we have instead chosen to map collective and nonlinear impacts of the environmental parameters on the expression of individual genes disregarding gene–gene interactions. While independence of genes (i.e. the lack of interactions) is biologically unrealistic, the analysis provides valuable insights into the interactive effects of environmental stressors on gene expression profiles, even though information related to gene epistasis is lost.

It is mathematically possible to allow all of the studied environmental variables to range over the observed values and generate an 18-dimensional picture of response for each gene. However, presenting such a hyper-dimensional space is not feasible, and we have chosen to use a variant of the Cannon & McKendry (2002). In Fig. 5, we present the predicted response of Hsp70 to temperature and pH variation for 4 models that were generated. As expected, our analysis confirms that Hsp70 is up-regulated by stressful conditions in oysters, especially by high temperatures and low pH. Our data also suggest that Hsp70 is more influenced by temperature than pH so that at a given pH, the temperature response is dramatic, but at a given temperature, the pH effect is weak. Thus, this analysis conforms to our expectations as to how a known stress-associated gene should respond to environmental conditions.

Data interpretation. In the southeastern US, particularly in Georgia and South Carolina, nearly all oyster beds are intertidal and oysters typically experience wide variation in temperature, salinity conditions, desiccation and oxygen availability. During the low tide, oysters are exposed to anaerobic conditions because of the closure of the shell, which prevents oxygen exchange with the environment. Hypoxia and anoxia during the shell closure in molluscs result in rapid and dramatic changes in the internal pH owing to the accumulation of CO_2 from aerobic respiration as well as acidic end products of anaerobic metabolism (Sokolova *et al.* 2000a,b; Michaelidis *et al.* 2005). There are a number of liabilities and benefits to this metabolic acidosis (Burnett & Milardo 2000). A moderate decrease in intracellular pH supports metabolic rate depression, which is the main adaptive mechanism to survive extreme stressors in intertidal molluscs (Brooks & Storey 1997; Sokolova *et al.* 2000a; Larade & Storey 2007). However, in extremely low pH (acidification), molluscs mobilize calcium carbonate stores in their connective tissues or the shell to buffer intracellular and extracellular pH (Sokolova *et al.* 2000b). Restoration of the acid–base homoeostasis can also contribute to the 'oxygen debt', a strong increase in oxygen demand during post-anoxic recovery (Ellington 1983; Vismann & Hagerman 1996; Lewis *et al.* 2007). Conceivably, variation in the sea water pH may affect intracellular pH regulation during metabolic acidosis and subsequent recovery either directly (via changes in extracellular or haemolymph pH that correlates with the environmental pH in oysters; Burnett 1997) or indirectly by affecting metabolic costs of acid–base regulation. Generally, acidic conditions (especially at pH <7.4) lead to reduced growth, higher mortality and shell abnormalities in bivalve molluscs, with the effect more evident in larvae and juveniles than adults (Calabrese & Davis 1966; Bamber 1990, Gazeau *et al.* 2007; Kurihara 2008; Kurihara *et al.* 2008; Parker *et al.* 2009). Thus, naturally low pH levels found at some sites in this study may exacerbate the physiological adjustments necessary for intertidal existence. Elevated temperatures may also cause physiological stress in eastern oysters, especially when the temperatures above 28 °C persist for a long time resulting in energy deficiency and cessation of growth (Surge *et al.* 2001; Lannig *et al.* 2006).

Transcriptomic responses to environmental pH and temperature discovered in this study corroborate these earlier physiological findings. Overall, our data indicate that under low pH and elevated temperature stress, the transcriptome responds by reducing the ATP-expensive protein synthesis and cell growth machinery and increasing ATP generation by elevating transcripts involved in the mitochondrial oxidative phosphorylation. Environmental temperature and pH also affected expression of genes involved in Ca signalling and homoeostasis and implicated in regulation of growth

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and energy metabolism. The response of four genes shown in Fig. 6 is an excellent illustration of how novel insights into oyster transcriptomic responses to stressful conditions can be obtained by this approach. For example, a gene involved in the protein synthesis machinery (rpL7) is more highly expressed in the near-optimal conditions of lower temperatures and higher pH (Fig. 6c). Elevated temperature and low pH lead to up-regulation of peroxinectin (Fig. 6a), a gene implicated in phagocytosis (Johanson *et al.* 1999), and a process important in clearing bacteria from haemolymph. As low pH and higher temperatures negatively impacts resistance to bacteria (Willson & Burnett 2000), it is possible the response of peroxinectin to these condition is related to defence response. The mitochondrial-encoded ND-2 gene (one of the subunits of a rate-limiting enzyme of ETC) is also up-regulated during exposure to low pH and elevated temperature, consistent with the higher energy demand of oysters under the stressful conditions. Paradoxically, cytochrome *b* transcripts, also involved in the ETC, appear to be down-regulated under low pH and high temperature stress in oysters (Fig. 6d). As the entire suite of mitochondrial genes are required for oxidative phosphorylation, it is odd that cytochrome *b* and ND-2 would exhibit such disparate response surfaces. At present, we have no explanation for the conundrum and can only speculate that this discrepancy may indicate either differential stability of the messages or differential regulation and/or activity of the mitochondrial gene promoters located on the light strain of mtDNA (from which cytochrome *b* is transcribed) vs. the heavy strand encoding most mitochondrial genes. Further studies are needed to determine the mechanisms and physiological implications of such differential responses of functionally related mitochondrial genes. However, an overall correspondence between the observed changes in gene expression (Fig. 6) with those that might be expected from physiological studies clearly indicates that transcriptomic data can be aligned to physiological outcomes. Another important conclusion from these findings is that the physiological responses to environmental conditions either elicit transcriptomic responses or are driven by transcriptomic responses to environmental signals, or both.

Finally, we attempted to summarize the response of many genes to pH and temperature by analysing the correlations between gene expression, pH and temperature (Fig. 7). While these linear correlations cannot reveal the subtleties shown in Fig. 6, they demonstrate an intriguing pattern, e.g. the genes that are positively correlated with temperature are negatively correlated with pH and *vice versa*. This may indicate that different suites of genes are positively associated with stress and others with the more benign, near-optimum conditions. It is also possible that parallel changes in gene expression patterns elicited by elevated temperatures and low pH may reflect the fact that in ectotherms such as oysters, high temperatures result in a decrease of intracellular pH, partially mimicking the effects of environmental acidification (Burton 2002). Notably, the negative correlations between gene responses to temperature and pH are considerably stronger in oyster gills than hepatopancreas (R^2 = 0.73 and 0.22 for gills and hepatopancreas, respectively). This may be attributed to the fact that gill tissues are directly exposed to the environment and thus would be expected to respond more strongly to the environmental change, whereas hepatopancreas is relatively more buffered by surrounding tissues particularly with regard to pH.

The strong dependence of most genes on temperature and pH is remarkable in yet another context. It is anticipated that oceanic temperature will rise (by approximately 5 °C) and pH will fall (by 0.3–0.7 units) over the next century (Carter *et al.*, 2007; Caldeira and Wickett, 2003). These global changes may exacerbate the pH challenges oysters already experience in the locations surveyed in this study. While these populations persist, our data indicate it comes at a cost and this cost in physiological terms can be directly related to gene expression. As such, transcript profiling may provide a relatively inexpensive means of assessing the impacts of climate change on physiological adaptation, provided of course that the linkage between transcripts and environment indicated in the present data continue to hold. At present, we do not know how general these results may be to other molluscs and if the oyster may serve as a model organism of the impact of climate change on bivalves. Further investigations using approaches similar to the one outlined in this study would be useful in determining the robustness of the observed responses in different species and their utility as integrated biomarkers of the environmental stress in marine molluscs.

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Supporting information

Additional supporting information may be found in the online version of this article:

Table S1 A listing of genes suggested to be important responders in gill (top 250 sensitivities) to the levels of the indicated environmental variables (see text for further explanation).

Table S2 A listing of genes suggested to be important responders in hepatopancreas (top 250 sensitivities) to the levels of the indicated environmental variables (see text for further explanation).

Table S3 A listing of gill genes that were important predictors (top 250 in sensitivities) of the difference between clusters of environmental variables.

Table S4 A listing of hepatopancreas genes that were important predictors (top 250 in sensitivities) of the difference between clusters of environ.

Table S5 Sensitivites of the indicatedgill gene to changes in the indicated enviromental variable and the associated average R2 of the training and Cv sets.

Table S6 Sensitivites of the indicated hepatopancreas gene to changes in the indicated enviromental variable and the associated average R2 of the training and Cv sets.

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